# Vector Spaces V: Bases, Dimension and Polynomial Spaces: Answer Key

Problem sheet for the Day 6 class · CSIR NET / GATE previous year questions

## Answers

| Q | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer | (A) | (C) | 15 | (C) | (C) | (B) | (A),(B) | (A),(D) | (A),(B),(C),(D) | (A),(B),(C),(D) |

## Hints and short solutions

1. **(A)** CSIR NET Dec 2016 $B_1, B_2$ give triangular matrices with non-zero diagonal (determinants 1 and 24). In $B_3$, the three vectors $(1,2,0,0), (2,1,0,0), (-5,5,0,0)$ lie in the 2-dimensional space $\mathbf{span}\{e_1, e_2\}$, so $B_3$ is dependent.
2. **(C)** GATE 2012 Need $f_i(v_j) = \delta_{ij}$ with $v_1 = (1,i), v_2 = (i,1)$. For (C): $f_1(v_1) = \frac{1}{2}(1+1) = 1$, $f_1(v_2) = \frac{1}{2}(i-i) = 0$, $f_2(v_1) = \frac{1}{2}(-i+i) = 0$, $f_2(v_2) = \frac{1}{2}(1+1) = 1$. In (A), $f_2(v_2) = i$.
3. **15** GATE 2013 $W_1 \cap W_2$ = polynomials vanishing at the 6 distinct points $1, \frac{1}{2}, 3, 4, 5, 7$, i.e. $p = (x-1)(x-\frac{1}{2})(x-3)(x-4)(x-5)(x-7)\,q$ with $\deg q \le 14$. So $\dim = 21 - 6 = 15$.
4. **(C)** GATE 2013 Rank 3 means the 3 columns in $\mathbb{F}_3^4$ are independent: $3^4 - 1$ choices for the first, $3^4 - 3$ outside its span, $3^4 - 3^2$ outside the plane of the first two. (Checked by brute force: 449280.)
5. **(C)** CSIR NET June 2016 $p^{(i)}(0) = i!\,a_i$. If $k \ge d$, $\|p\|_k = 0$ forces every coefficient to vanish, so $p = 0$; the other norm axioms always hold. If $k < d$, then $\|x^d\|_k = 0$ with $x^d \neq 0$.
6. **(B)** CSIR NET June 2016 $A$ is strictly upper triangular ($T(x^j) = jx^{j-1}$): trace 0, $\det A = 0$, $A^{11} = 0$, all eigenvalues 0.
7. **(A),(B)** CSIR NET June 2016 $Tp$ is the constant polynomial $p'(1)$; it is non-zero for $p = x$, so the range is the constants (dimension 1) and by rank–nullity $\dim \operatorname{Ker} T = (n+1) - 1 = n$.
8. **(A),(D)** CSIR NET Dec 2017 $D^4 = 0$ but $D^3 x^3 = 6 \neq 0$: nilpotent of index 4 = $\dim V$, so a single $4 \times 4$ Jordan block. Rank $= \dim D(V) = \dim P_2 = 3$; a non-zero nilpotent is not diagonalizable.
9. **(A),(B),(C),(D)** CSIR NET Dec 2015 $\operatorname{Range} T^2 \subseteq \operatorname{Range} T$ and $\operatorname{Ker} T \subseteq \operatorname{Ker} T^2$ with equal dimensions (rank–nullity), so both are equalities. If $v = Tw$ and $Tv = 0$, then $w \in \operatorname{Ker} T^2 = \operatorname{Ker} T$, so $v = 0$. (D) is (C) rewritten using (A),(B).
10. **(A),(B),(C),(D)** CSIR NET Dec 2015 $T$ reverses the coefficient vector, so $T^2 = I$: $T$ is bijective. Its matrix in the basis $\{1, x, \ldots, x^n\}$ is a permutation matrix, so $\det T = \pm 1$ (in fact $(-1)^{\lfloor (n+1)/2 \rfloor}$).